National EUDI Wallet Project · Sandbox Use Case

EUDI Wallet Integration into an EHDS-aligned Health Dataspace

Use Case: BMDS Deutschland Architektur (Identitäten)

EHDS Regulation | eIDAS 2.0 / EUDI Wallet | Eclipse DCP 1.0 | Eclipse Dataspace Components | FHIR R4 · OMOP CDM

Use Case at a Glance

Today we operate an EHDS-aligned health dataspace based on Eclipse Dataspace Components (EDC) and the Decentralized Claims Protocol (DCP 1.0). Credentials are exchanged between organisational participants via **Business Wallets**. The missing piece is a citizen-controlled wallet for identity, health-data access rights, and consent. The **EUDI Wallet** closes this gap, enabling patient-controlled access (EHDS Art. 3) and consent-driven secondary use (EHDS Art. 10) end-to-end.

PRIMARY ROLE
Service Provider / Technical Integrator (Issuer + Relying Party)

INDUSTRY
Healthcare · Public Sector

LEGAL BASIS
EHDS Regulation · eIDAS 2.0 · GDPR (Art. 6, 9, 15, 17, 20)

CREDENTIALS
PID · Health-data EAAs · Consent Attestations

TECH STACK
EDC · DCP 1.0 · Neo4j · Verifiable Credentials

STATUS
Demo & Integration Platform live · EUDI integration: next phase

DOCUMENT CONTENTS

1. Roles & Actors — who issues, who holds, who verifies
2. Architecture — EUDI Wallet inside the EDC / DCP dataspace
3. User Journey — walked through with live demo screenshots
4. EHDS Compliance & Article Mapping

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1 · Roles & Actors

The use case follows the classic *Issuer – Holder – Verifier* trust triangle from eIDAS 2.0 and the W3C Verifiable Credentials model, extended with a *Service Provider* role for the technical integration of the EUDI Wallet into the EDC-based dataspace.

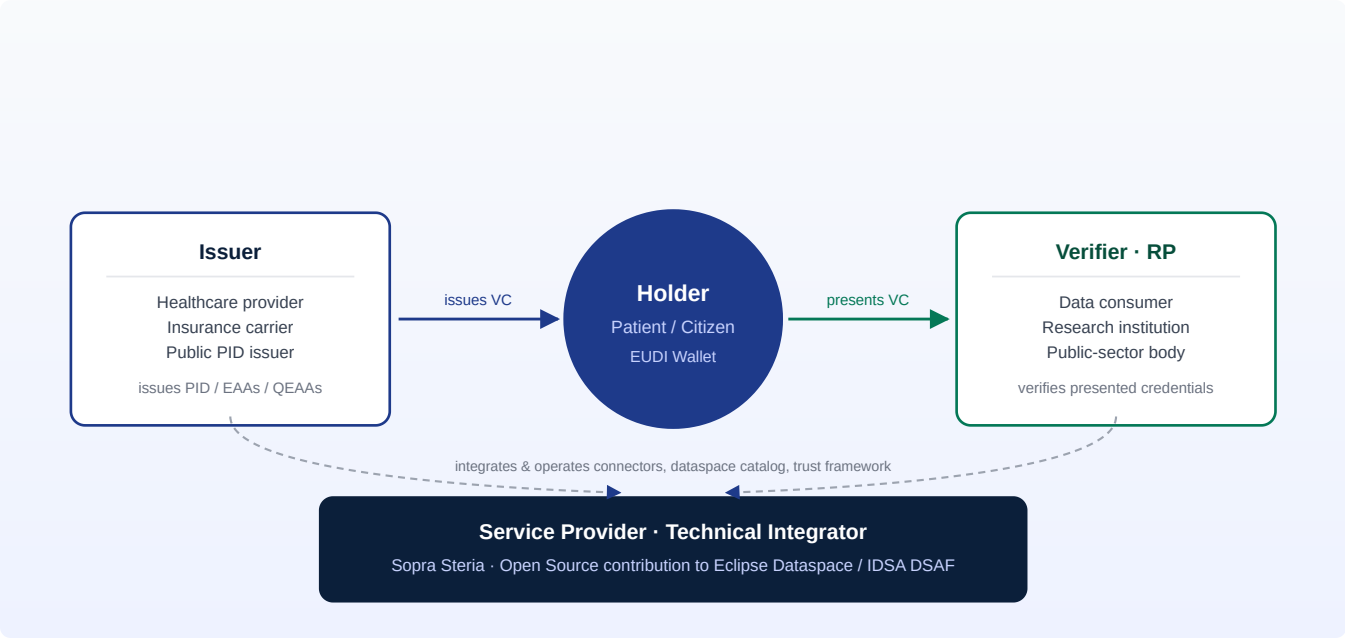


Figure 1 — Roles in the EUDI Wallet × EDC dataspace integration.

ROLE MAPPING

Role	Actor	Responsibility in this use case
Issuer	Healthcare providers, statutory health insurers, public PID issuer	Issue PID, health-data EAAs (access rights, professional roles), consent attestations into the patient's EUDI Wallet and into Business Wallets at EDC connectors.
Holder	Patient / Citizen	Holds credentials in the EUDI Wallet, presents selectively, grants and revokes consent.
Verifier / RP	Data consumers, research institutions, public-sector bodies	Verify credentials presented by the patient or via the EDC connector before granting access to health data.
Service Provider	Sopra Steria + Open Source community	Integrate the EUDI Wallet with EDC + DCP 1.0; operate the demo & integration platform; align with EHDS & eIDAS 2.0 standards.
Intermediary	Federated catalog, trust anchors	Provide discovery and trust establishment between participants (out of scope for sandbox MVP, planned for next iteration).

2 · Architecture

The architecture combines the **citizen-side EUDI Wallet** with the **organisation-side Business Wallet** already implemented inside EDC connectors via DCP 1.0. Both wallets exchange Verifiable Credentials over the Dataspace Protocol (DSP), governed by a federated catalog and trust framework.

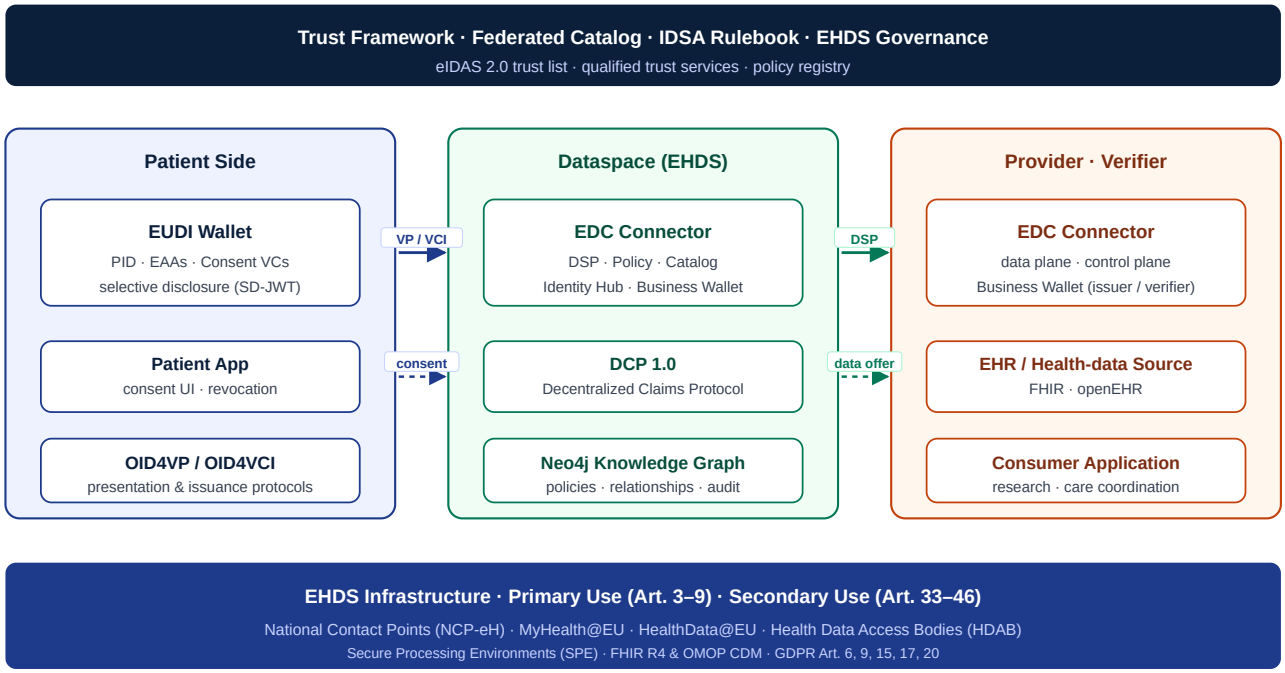


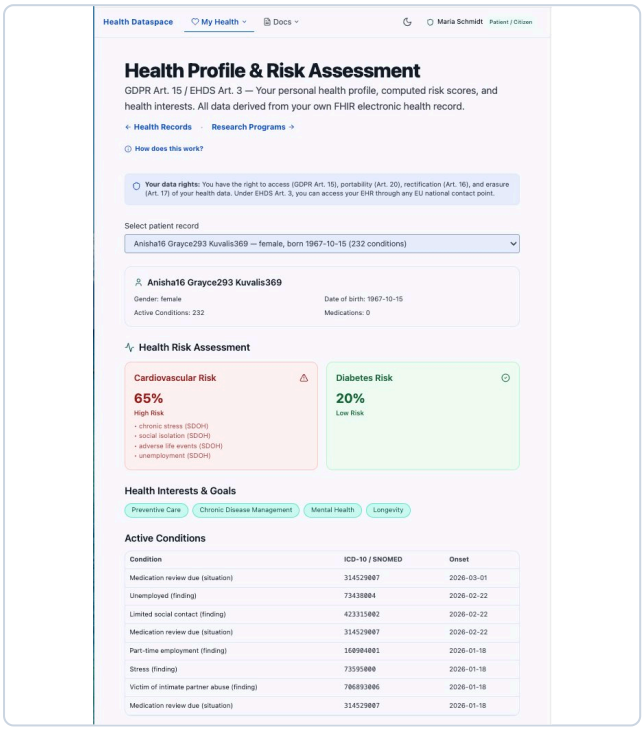
Figure 2 — Layered architecture: trust framework (top), three-zone runtime (middle), EHDS infrastructure foundation (bottom).

COMPONENT RESPONSIBILITIES

- **EUDI Wallet (patient):** stores PID and health-related EAAs; performs selective disclosure via OID4VP; generates consent VCs.
- **EDC Connector + Business Wallet:** negotiates contracts (DSP), enforces usage policies, holds organisational credentials via DCP 1.0.
- **Decentralized Claims Protocol (DCP 1.0):** binds participant identity to dataspace messages without a central IAM.
- **Knowledge graph (Neo4j):** stores policy relationships, audit trail, and dataspace topology.
- **Trust framework:** eIDAS 2.0 trust list, IDSA Rulebook, EHDS governance — all federated.

3 · User Journey — Patient Consent & Data Access

The journey is shown through screenshots from the live Minimum Viable Health Dataspace demo. Each step maps a patient action to an EHDS / GDPR right and to the credential the EUDI Wallet will carry once integrated. Steps 1–2 cover patient access and exploration; steps 3–4 (next page) cover consent and the research-insights loop.

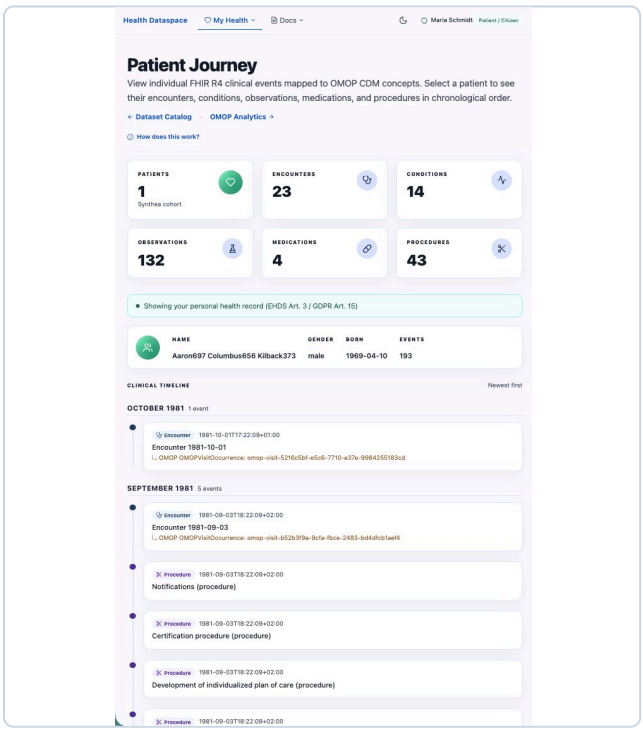


1 Access & Risk Assessment

EHDS Art. 3 | GDPR Art. 15

The patient (Maria Schmidt) authenticates and views her own electronic health record: cardiovascular and diabetes risk scores, health interests, and active conditions. All scores are derived locally from her FHIR record — nothing leaves the patient context yet.

EUDI Wallet adds: authentication via PID (selectively disclosed) replaces platform login. Her presence proof becomes portable across EU member states via the eIDAS 2.0 trust list.



2 Explore Clinical Timeline

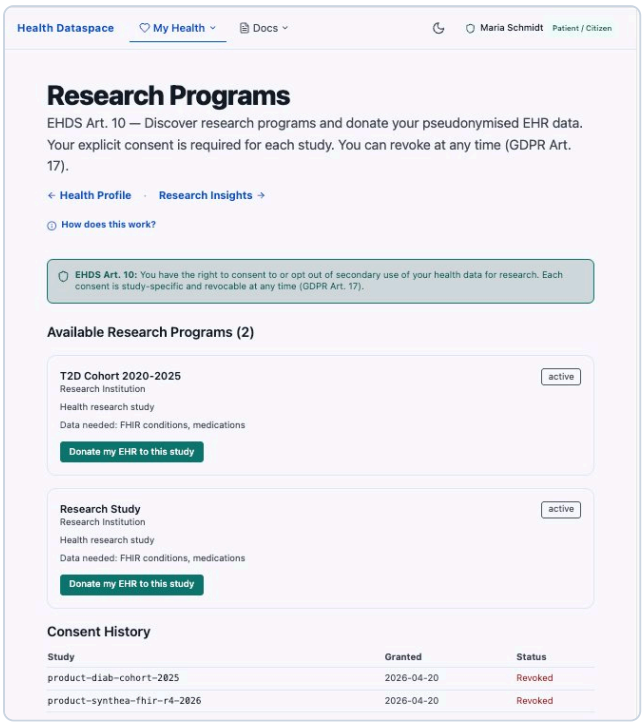
EHDS Art. 5 | FHIR R4 → OMOP CDM

The patient drills into individual clinical events — encounters, conditions, observations, medications, procedures — mapped from FHIR R4 to OMOP CDM concepts. This is the granular data model that any subsequent consent decision will scope.

EUDI Wallet adds: the wallet stores a health-data *access EAA* (Electronic Attribute Attestation) issued by her statutory insurer, proving her right to the underlying record without exposing it.

3 · User Journey — Consent & Research Loop

Steps 3–4 close the loop: the patient grants explicit, study-specific consent for secondary use, and later receives aggregate insights back. The EUDI Wallet is the trust anchor that makes the consent portable, auditable, and revocable.

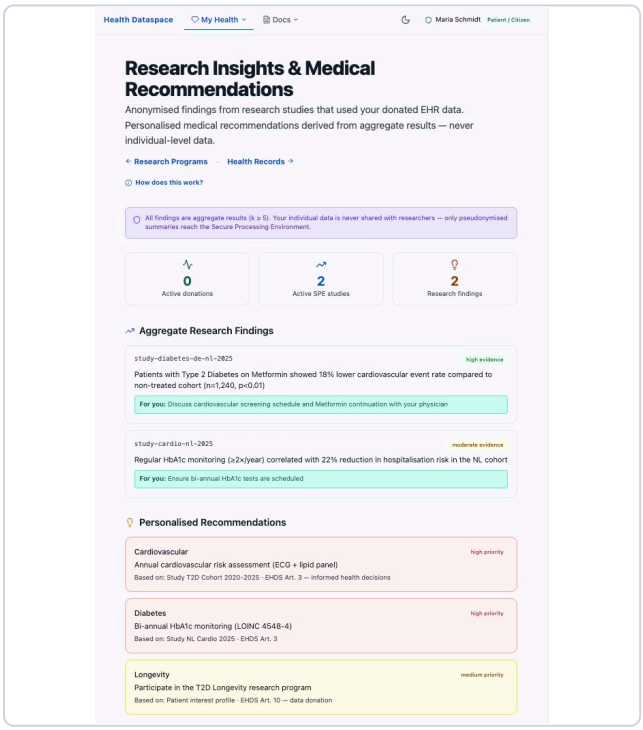


3 Grant Consent · Donate EHR

EHDS Art. 10 | GDPR Art. 17

The patient sees available research programs, reviews scope and data needs (FHIR conditions, medications), and grants consent per study. The Consent History panel shows that previous donations were revoked — proof that revocation works end-to-end.

EUDI Wallet adds: instead of a server-side flag, consent becomes a **signed Verifiable Credential** issued from the wallet, presented to the EDC connector via OID4VP, and revocable any time through a VC status list that propagates to all verifiers.



4 Receive Aggregate Insights

EHDS Art. 50 | k ≥ 5 anonymity

The loop closes. The patient receives anonymised findings from studies that used her donated data — never individual-level results. Personalised recommendations (cardiovascular screening, bi-annual HbA1c) are derived locally from the aggregates.

EUDI Wallet adds: the patient's individual record never leaves the Secure Processing Environment (SPE). The wallet's consent VC binds the purpose, retention, and aggregation threshold; the EDC connector enforces the policy at egress.

Behind the scenes — between steps 3 and 4, the consumer-side EDC connector negotiates a contract via the Dataspace Protocol (DSP), verifies the patient's consent VC via DCP 1.0, retrieves the scoped FHIR subset, and runs the analysis inside the SPE. Only aggregate results (k ≥ 5) flow back to the patient.

4 · EHDS Compliance & Article Mapping

The European Health Data Space (Regulation (EU) 2025/327) defines two parallel pathways: **primary use** (the patient and their care team accessing the record) and **secondary use** (research, policy, innovation, regulatory activities). The EUDI Wallet is the trust anchor for both — it carries the patient's PID, EHR access rights, and explicit, revocable consent.

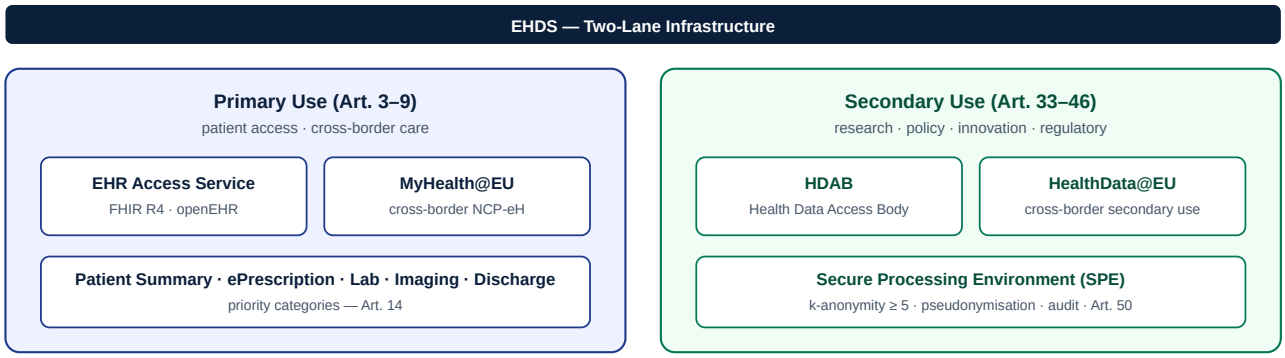


Figure 4 — EHDS two-lane infrastructure: primary use (left) for patients and care teams, secondary use (right) for research and policy.

ARTICLE-BY-ARTICLE MAPPING

EHDS / GDPR	What it requires	How this use case satisfies it
EHDS Art. 3	Right to access one's electronic health data	Patient views her full FHIR record in the My Health area; all access events are logged.
EHDS Art. 8b	Right to rectification, erasure, restriction	Patient app surfaces GDPR Art. 16/17/18 controls; revocation propagates via VC status list.
EHDS Art. 10	Secondary use — opt-in / opt-out, study-specific	Per-study consent VCs issued from the EUDI Wallet; revocable any time (see screenshots p. 6).
EHDS Art. 14	Cross-border interoperability via MyHealth@EU	EDC connector exposes a national-contact-point-compatible interface; PID is eIDAS 2.0 trust-listed.
EHDS Art. 50	Secure Processing Environment for secondary use	Aggregate findings only (k ≥ 5); individual records stay in the SPE; audit on every query.
GDPR Art. 6 + 9	Lawful basis for processing health data	EUDI-Wallet-issued consent VC binds purpose, scope, retention; verifiable at the EDC connector.

NEXT STEPS

- Onboard to the SPRIND EUDI Wallet sandbox; obtain test PID issuer + verifier endpoints.
- Extend the Minimum Viable Health Dataspace with an OID4VP / OID4VCI bridge to the EDC Identity Hub.
- Validate consent VC schema with EHDS governance (HDAB) and national health-data authorities.
- Publish a reference integration as open-source contribution to Eclipse Dataspace & IDSA DSAF.

References — EHDS Reg. (EU) 2025/327 · eIDAS 2.0 (Reg. (EU) 2024/1183) · Eclipse Dataspace WG · IDSA DSAF · github.com/ma3u/MinimumViableHealthDataspacev2